
Dancing with the Stars (DWTS) determines weekly eliminations by aggregating judges' scores and audience votes under different voting rules across seasons. While this hybrid mechanism is intended to balance technical merit and popularity, it has repeatedly generated controversial outcomes, raising concerns about fairness, stability, and potential bias toward fan voting.

In this paper, we develop a unified quantitative framework to analyze and evaluate the DWTS voting system. First, we formulate the estimation of audience vote shares as a constrained Bayesian inverse problem. Because fan votes are unobserved, we infer their posterior distributions using Markov Chain Monte Carlo (MCMC) sampling, exploiting elimination outcomes as hard inequality constraints and imposing temporal smoothness priors across weeks. This approach enables probabilistic reconstruction of latent fan vote dynamics and provides uncertainty quantification at the contestant-week level.

Second, using posterior samples of fan votes, we conduct conditional simulations under the two historically used aggregation rules—the Rank-based method and the Percentage-based method. We compare their stability, bias toward fan influence, and sensitivity to posterior uncertainty across seasons, with particular attention to controversial contestants and the impact of the Judges' Save rule.

Third, we introduce a dual-track hierarchical mixed-effects model to disentangle the mechanisms governing judges' scores and fan votes. By modeling professional partner effects and celebrity characteristics separately for each track, we quantify how technical evaluation and audience preference respond differently to the same covariates.

Finally, based on these analyses, we propose a Progressive Technical Fairness System (PTFS), a dynamic weighting scheme that adjusts the balance between judges' scores and fan votes over the course of a season. Our results demonstrate that percentage-based aggregation is generally more robust to fan-vote uncertainty, while PTFS further improves fairness and stability. The proposed framework provides actionable insights for DWTS and offers a general methodology for designing fair voting systems in entertainment competitions.

Bayesian inference, Latent vote estimation, Voting aggregation rules, Fairness analysis, Mixed-effects models

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Dancing with the Stars (DWTS) is a televised dance competition where celebrities, paired with professional dancers, are evaluated weekly by expert judges and audience votes. The hybrid scoring mechanism—combining technical assessment with popular preference—has generated recurring controversies: contestants with persistently low judge scores have advanced or won through fan support alone (e.g., Bobby Bones in Season 27). These outcomes raise fundamental questions about fairness, stability, and the appropriate balance between expert evaluation and public participation in entertainment competitions.

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Task 1: Develop a model to estimate the distribution of fan votes, evaluate the consistency and uncertainty of the estimates, and visualize the results.
- Task 2: Using the estimated fan votes and the remaining competition data:
 - , compare the outcomes of the rank-based and percentage-based vote aggregation methods across seasons and analyze the degree to which each method favors fan influence.
 - , examine the effects of different aggregation methods and the rule allowing judges to select the eliminated contestant from the bottom two on the outcomes of controversial contestants.
 - , based on the analysis, recommend an appropriate vote aggregation method and determine whether the judges' selection rule should be retained for future seasons.
- Task 3: Develop a model to analyze the impact of professional dancers and celebrity characteristics (such as age and industry) on competition performance, and compare whether these factors affect judges' scores and fan votes in the same way.
- Task 4: Propose a new weekly competition system that combines fan votes and judges' scores, justify why it is more fair or superior in other aspects, and explain why it should be adopted by the show producers.

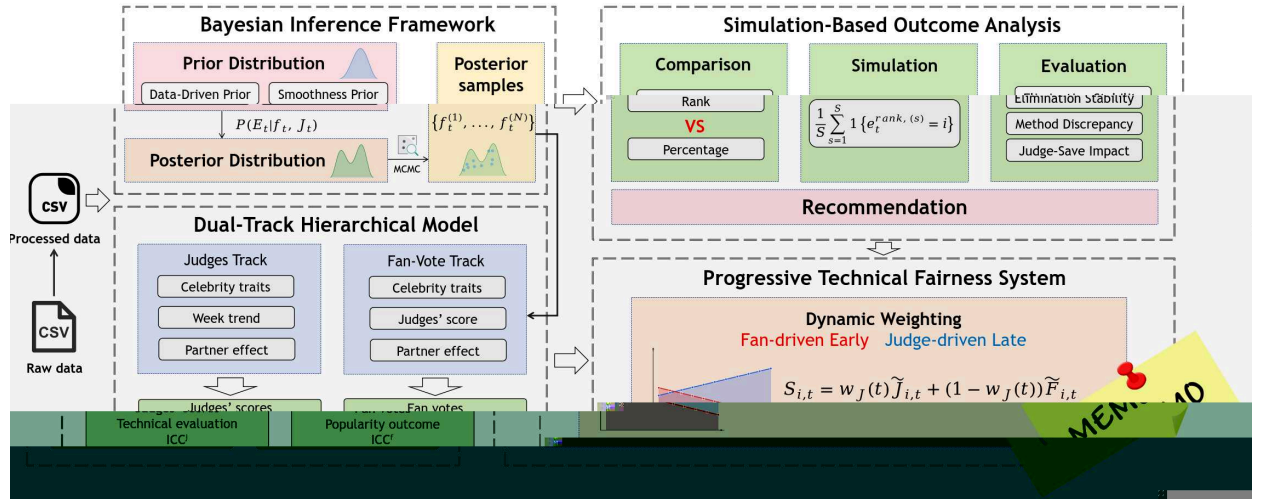


Figure 1: The workflow of our work.

Our approach systematically addresses the four tasks through interconnected analytical frameworks (Figure 1). First, in the [Bayesian Inference Framework](#) [1], we estimate latent fan-vote distributions from elimination outcomes using MCMC sampling[2] with data-driven priors. Second, in the [Simulation-Based Outcome Analysis](#), we compare rank-based and percentage-based aggregation methods across seasons to evaluate their impact on elimination stability and fairness. Third, in the [Dual-Track Hierarchical Model](#) [3], we separately model judges' scores and fan votes to quantify how professional partners and celebrity characteristics differentially affect technical evaluation versus popularity outcomes. Finally, the [Progressive Technical Fairness System](#) (PTFS) proposes a dynamic weighting scheme derived from social choice theory[4] that progressively adjusts the balance between judges' influence and fan votes throughout the season.

Fan votes are unobserved. The weekly elimination order is mechanically determined by the official aggregation rule applied to judges' scores and fan votes. Therefore, observed elimination outcomes impose inequality constraints on the latent fan vote shares, which are exploited for identification and posterior inference in Task 1.

A contestant's fan vote share is assumed to change gradually rather than exhibit abrupt structural jumps from one week to the next within a season. We encode this assumption through a temporal smoothing prior in Task 1, which regularizes the posterior distribution and renders the inverse problem well-posed.

Judges' scores primarily reflect technical evaluation, whereas fan votes capture popularity and viewer preference. Consequently, we model the two outcomes using separate tracks in Task 3 and examine how professional partners and celebrity attributes differentially affect each mechanism.

The dataset comprises weekly performance records, individual judge scores, season outcomes, and professional–celebrity partnership information. Over the course of

certain institutional features—such as the size of the judging panel, elimination rules, and score aggregation procedures—have varied across seasons. While the core data structure remains stable, these design changes introduce systematic differences that must be accounted for prior to modeling. Accordingly, data preprocessing emphasizes institutional sources of missingness and variation that are inherent to the competition format.

Missing judge scores arise from clearly defined structural mechanisms rather than random loss. As shown in Table 1, a portion of score fields is unused in seasons or weeks with fewer judges, while additional missing values occur mechanically after contestant elimination, when no further weekly scores are generated.

Table 1: Sources of missing values in judge score records.

Judge slot not used	Fewer judges in certain seasons or weeks	2,499
Post-elimination	No scores recorded after contestant elimination	6,904
Total score cells		18,524

In addition to missingness, the elimination mechanism itself varies across seasons. While most weeks involve a single elimination, some feature double eliminations, and a small number involve three contestants eliminated simultaneously. Figure 2 summarizes, for each competition week, the number of seasons exhibiting different elimination counts. This variation implies that elimination outcomes cannot be inferred from score rankings alone without conditioning on the contemporaneous rules.

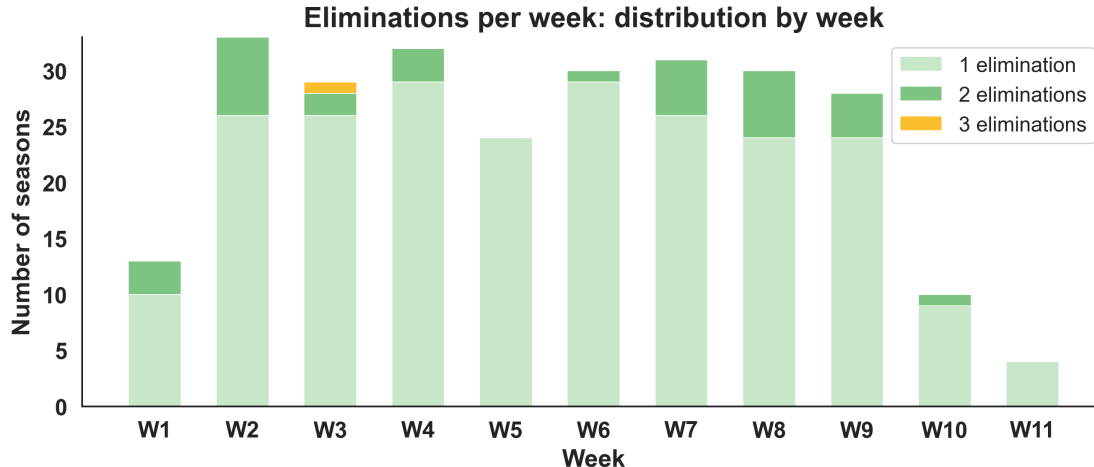


Figure 2: Distribution of elimination counts per week across seasons.

Observed score values at the boundaries of the empirical range are retained and interpreted within their institutional context. Zero scores correspond to post-elimination records, while values exceeding standard score limits arise from competition-specific bonus rules. Exceptional events such as contestant withdrawals are encoded through indicator variables to preserve the structural integrity of the dataset.

Following data cleaning, the raw score data were restructured to support time-series analysis. The original wide-format table, in which scores from different weeks and judges were stored in separate columns, was unsuitable for temporal modeling. Accordingly, the data were transformed

into a long-format structure, where each record represents a combination (Table 2).

Table 2: Long-format score schema: one row per contestant–week–judge combination.

contestant_id	Contestant identifier (e.g. celebrity name + season)
season	Season number
week	Competition week index
judge	Judge index within the week
score	Judge score for that contestant–week–judge

Based on the long-format data, weekly aggregation was performed. After excluding zero-score entries generated after elimination, scores were grouped by to compute weekly total and average scores. Within each season, contestants were ranked by weekly total score to obtain week-level rankings.

To integrate information across time scales, season-level statistics were constructed from the weekly summaries. These include the number of weeks competed, total season score, average weekly score, and average rank. The weekly data, season-level statistics, and partner information were then merged to form a comprehensive dataset containing temporal, performance, and contextual variables.

After data transformation, features were recombined from multiple temporal scales: (1) features capturing short-term dynamics (previous score, week-to-week change, deviation from cumulative average); (2) statistics describing long-term performance (total score, average rank, score volatility); and (3) features summarizing professional dancer competitiveness (championships, top-3 finishes, average judge score). Categorical variables such as industry were encoded into model-compatible formats (Table 3).

Table 3: Engineered features across temporal scales.

Weekly	prev_week_score	Previous week total score
	score_change	Week-to-week score change
	deviation_from_avg	Deviation from cumulative average
Season	weeks_competed	Number of weeks competed
	season_avg_weekly_score	Season average weekly score
	score_volatility	Score standard deviation
	avg_rank	Average weekly rank
Partner	championships	Partner's championship count
	top3_finishes	Partner's top-3 finish count
	avg_judge_score	Partner's historical average score
Other	industry_code	Celebrity industry category
	is_controversy_case	Binary indicator for controversy

The fundamental challenge in analyzing DWTS voting dynamics is that actual fan votes are never disclosed. We formulate this as a constrained Bayesian inverse problem[5]. Let $C_t = \{c_1, \dots, c_n\}$ denote the set of n contestants competing in week t . For each contestant i , define $J_i \in [0, 100]$ as the total judges' score, $E_t \subset C_t$ as the observed eliminated set, and $f_{i,t} \in (0, 1)$ as the latent fan vote share satisfying the simplex constraint $\sum_i f_{i,t} = 1$. The fan vote vector $\mathbf{f}_t = (f_{1,t}, \dots, f_{n,t})$ is not directly observable, and elimination outcomes provide only limited ranking information. We therefore seek the posterior distribution

$$P(\mathbf{f}_t \mid E_t, \mathbf{J}_t) \propto P(E_t \mid \mathbf{f}_t, \mathbf{J}_t) \cdot P(\mathbf{f}_t \mid \mathbf{f}_{t-1}) \cdot P(\mathbf{f}_t) \quad (1)$$

where the three terms correspond to the likelihood, temporal prior, and static prior, respectively.

The likelihood encodes elimination rules as hard constraints. DWTS has employed two aggregation schemes: the rank-based method (Seasons 1–2 and 28–34), where $\text{Score}_i = \text{Rank}(J_i) + \text{Rank}(f_{i,t})$, and the percentage-based method (Seasons 3–27), where $\text{Score}_i = \frac{J_i}{\sum_j J_j} + f_{i,t}$. Let $k_t = |E_t|$ denote the number of eliminations in week t . For standard weeks (Seasons 1–27), the likelihood is $P(E_t \mid \mathbf{f}_t, \mathbf{J}_t) = \mathbb{I}(\text{Bottom} - k_t = E_t)$. From Season 28 onward, the Judges' Save rule allows judges to choose whom to eliminate from the bottom two, so the constraint relaxes to requiring E_t be contained in that set.

The prior distribution $f_{i,t} \sim \mathcal{N}(\mu_i, \sigma_{\text{prior}}^2)$ incorporates industry-specific base rates derived from historical survival analysis ($\sigma_{\text{prior}} = 0.25$), with an age adjustment factor $\mu_i \leftarrow \mu_i \cdot \left(0.9 + 0.1 \cdot \exp\left(-\frac{(\text{Age}_i - 40)^2}{30^2}\right)\right)$ that peaks at age 40. A temporal smoothing prior $P(\mathbf{f}_t \mid \mathbf{f}_{t-1}) \propto \exp\left(-\lambda \sum_i (f_{i,t} - f_{i,t-1})^2\right)$ with $\lambda = 0.5$ regularizes week-to-week changes.

Combining the static prior and temporal smoothness terms, the posterior log-probability can be written as

$$\log P(\mathbf{f}_t) \propto -\frac{1}{2\sigma_{\text{prior}}^2} \sum_i (f_{i,t} - \mu_i)^2 - \lambda \sum_i (f_{i,t} - f_{i,t-1})^2 \quad (2)$$

subject to the simplex constraint and the elimination-consistency condition. If the likelihood constraint is violated, we set

$$\log P(\mathbf{f}_t) = -\infty \quad (3)$$

Since the posterior has no closed form, we employ adaptive Metropolis-Hastings sampling[1] with log-space proposals $\log(\mathbf{f}^*) = \log(\mathbf{f}^{(k)}) + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma_{\text{prop}}^2)$, followed by renormalization. The acceptance probability $\alpha = \min\left(1, \frac{P(\mathbf{f}^*)}{P(\mathbf{f}^{(k)})}\right)$ is used, with an adaptive mechanism targeting an acceptance rate of approximately 0.35. Thinning (every 5th sample retained) reduces autocorrelation.

Figure 3 shows the distribution of estimated fan vote shares for eight selected seasons, illustrating how inferred shares concentrate at low values when many contestants remain and spread toward higher values as the field narrows.

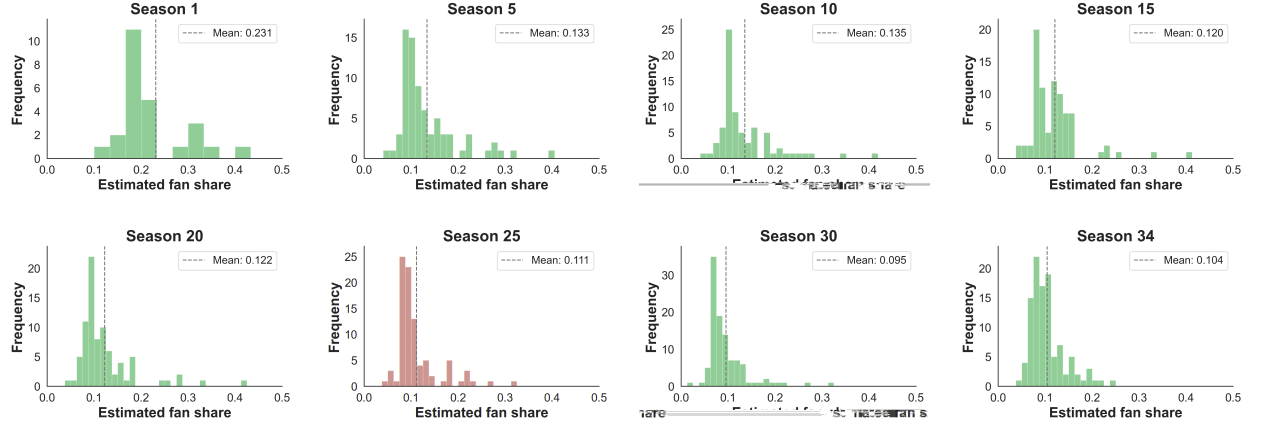


Figure 3: Distribution of estimated fan vote shares.

We assess model performance using four complementary metrics. Elimination accuracy measures the proportion of weeks where the predicted bottom- k_t set exactly matches the observed eliminations: $\text{Acc} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\text{Bottom} - k_t(\hat{f}_t, J_t) = E_t)$. Rank correlation, measured by Kendall's τ [6], quantifies agreement between predicted and actual final rankings. Posterior consistency reflects how concentrated the posterior is: $\text{Consistency}_t = \frac{1}{N} \sum_{m=1}^N \mathbb{I}(\arg \min_i \text{Combined}_i^{(m)} \in E_t)$. Finally, the effective sample size[1] assesses chain mixing.

$$\text{Acc} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\text{Bottom} - k_t(\hat{f}_t, J_t) = E_t) \quad (4)$$

The agreement between the predicted ranking and the actual ranking is measured by Kendall's τ [6] (rank correlation),

$$\tau = \frac{2(C - D)}{n(n - 1)}, \quad (5)$$

where C is the number of concordant pairs (same order in both rankings), D is the number of discordant pairs, and n is the number of ranked contestants. Values of τ closer to 1 indicate stronger agreement between predicted and actual rankings.

For each elimination week, the proportion of MCMC samples for which the predicted lowest combined score falls within the actual eliminated set E_t ; it reflects how concentrated the posterior is for that week.

$$\text{Consistency}_t = \frac{1}{N} \sum_{m=1}^N \mathbb{I}(\arg \min_i \text{Combined}_i^{(m)} \in E_t) \quad (6)$$

The number of effectively independent draws in the chain; low ESS indicates strong autocorrelation and suggests that posterior intervals should be interpreted with caution. [1], [7]

$$\text{ESS} = \frac{N}{1 + 2 \sum_{h=1}^{\infty} \rho_h} \quad (7)$$

where ρ_h denotes the autocorrelation coefficient at lag h .

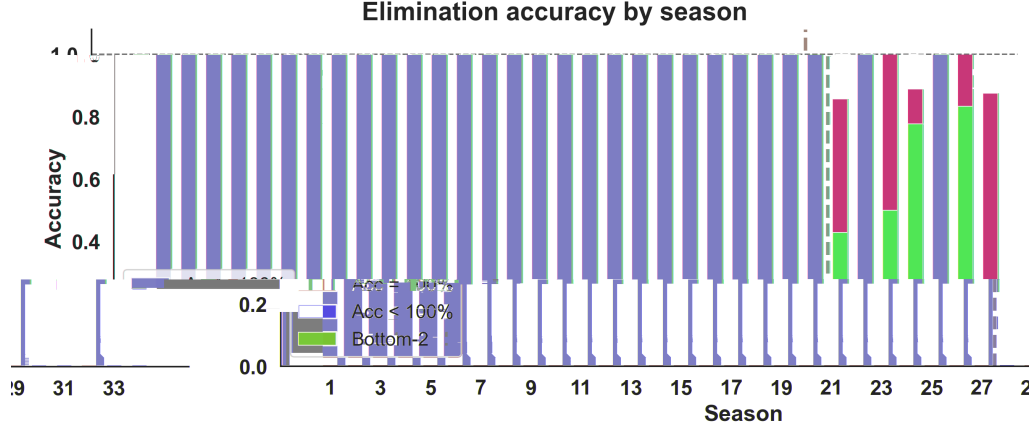


Figure 4: Elimination accuracy by season.

The model achieves an elimination accuracy of 93.49% (244 out of 261 elimination weeks), correctly predicting which contestants would be eliminated under the inferred fan vote distribution (Fig. Figure 4). The predicted final rankings exhibit strong agreement with observed outcomes, with Kendall’s $\tau = 0.994$ across all seasons; the model correctly identifies the champion in all 34 seasons and recovers the entire top-three ordering in 100% of cases. Posterior consistency averages 0.90 (std. 0.22), and the mean effective sample size of 191 (minimum 116) confirms adequate chain mixing.

Using the posterior fan-vote distribution from Task 1 and the observed judges’ scores, we simulate each week under both aggregation rules: the Rank-based method (combined score = rank from judges + rank from fan votes) and the Percentage-based method (combined score = normalized judges’ score + fan share). For every contestant-week, we obtain fan shares from the MCMC posterior and combine them with the actual judge scores to compute rank-sum and percentage-sum; the contestant(s) with the lowest combined score in that week are taken as eliminated under the corresponding rule. This yields, for each week with elimination, two (possibly different) eliminated contestants—one under Rank and one under Percentage. We then use two indicators to compare the two methods: whether they disagree on who is eliminated (DiffProb), and which rule is more biased toward fan voting (Judge-Favorite Elimination Gap, JFEG).

We define the probability that the two methods disagree on the eliminated contestant in week t as

$$\text{DiffProb}_t = \frac{1}{S} \sum_{s=1}^S \mathbf{1}\{e_t^{\text{rank},(s)} \neq e_t^{\text{pct},(s)}\} \quad (8)$$

where $e_t^{\text{rank},(s)}$ and $e_t^{\text{pct},(s)}$ denote the contestant eliminated in week t under the Rank and Percentage rules in simulation s , and S is the number of (posterior or bootstrap) runs. With a single run per week, $\text{DiffProb}_t = 1$ if the two rules eliminate different contestants and 0 otherwise. A higher mean DiffProb indicates that the choice of aggregation rule more often changes who is eliminated.

To assess which rule is more biased toward fan voting, we use a subgroup-based measure grounded in conditional expectation. Define the subgroup

H_t as contestants who are (top q_J in judge rank) and (bottom q_F in fan rank) in week t —i.e., judge favorites that fans do not like. Let

$$\bar{P}_t^m = \left(\frac{1}{|H_t|} \right) \sum_{i \in H_t} P(\text{eliminated} = i \mid \text{method} = m) \quad (9)$$

be the elimination probability in H_t under method m (with a single realization per week, $\bar{P}_t^m$ is $\frac{1}{|H_t|}$ if the contestant eliminated under m lies in H_t , and 0 otherwise). The in week t is

$$\Delta_t^{\text{JFEG}} = \bar{P}_t^{\text{pct}} - \bar{P}_t^{\text{rank}}. \quad (10)$$

If $\Delta_t^{\text{JFEG}} > 0$, the Percentage method eliminates judge favorites (that fans dislike) more often than the Rank method, so we interpret the Percentage method as in that week; if $\Delta_t^{\text{JFEG}} < 0$, the Rank method is more fan-biased. This comparison is restricted to (weeks where the two methods eliminate different contestants), so that the bias measure is informative. Season-level conclusions are obtained by averaging Δ_t^{JFEG} over disagreement weeks.

Across 295 weeks with elimination, the two methods yield different eliminated contestants in 30 weeks, giving a mean DiffProb of . Thus in about 90% of weeks the Rank and Percentage rules agree on who is eliminated; in the remaining 10% the choice of rule changes the outcome. On the 30 disagreement weeks, we compute JFEG for the 20 weeks in which the subgroup H_t (judge top 30%, fan bottom 30%) is non-empty. The mean Δ^{JFEG} over these weeks is (positive). Hence on weeks where the two rules actually disagree, the Percentage method eliminates judge favorites that fans dislike more often than the Rank method—i.e., the Percentage method is in the sense of JFEG. In sum: (1) the aggregation rule has a material effect in roughly one in ten elimination weeks; (2) when it does, the Percentage-based rule tends to favor fan preference over judge preference more than the Rank-based rule.

In controversial cases (such as Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones), we examine two questions: (1) whether the choice of aggregation method leads to the same result for each contestant; (2) how adding the Judges' Save rule (judges choose which of the bottom two pairs to eliminate) affects outcomes. We present the first part here and the second in the following subsection.

Using the inferred fan votes and the same simulation setup as in Task 2, we compute each contestant's weekly rank under both the Rank-based and Percentage-based methods. The grey shaded region in each panel is the : in any week, a contestant whose rank falls at or below the cutoff (the dark grey line) would be eliminated under a strict “touch = out” rule; we truncate each series at the first week it touches the zone so that the plot reflects the moment of (simulated) elimination. Figure 5 shows the rank trajectories for the four controversial contestants.

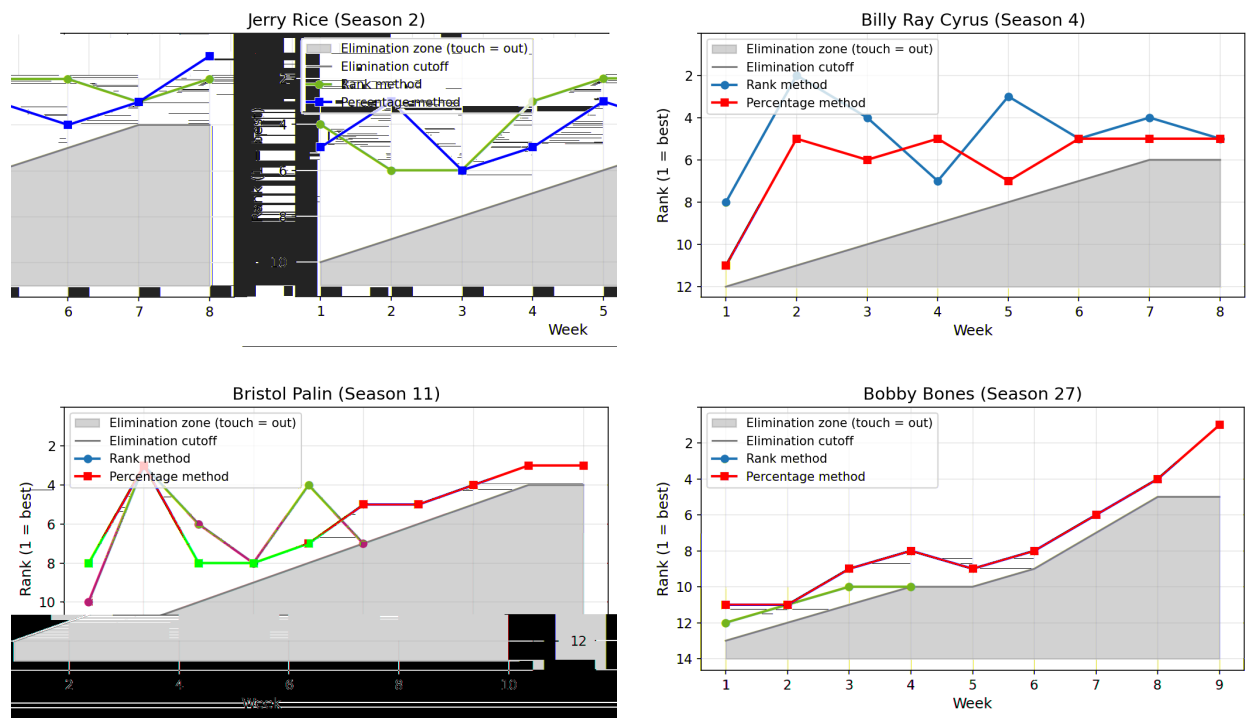


Figure 5: Weekly rank under the Rank method and the Percentage method.

The two methods do yield the same outcome for every contestant.

- and : Under both the Rank and Percentage methods, their ranks stay above the elimination cutoff for all weeks shown. Week-by-week ranks differ between the two methods, but neither would be eliminated in the simulated period.
- : Under the Rank method, her rank touches the cutoff in Week 6, so she would be eliminated in that week; the blue series stops there. Under the Percentage method, her rank remains above the cutoff through Week 10, so she would not be eliminated. Thus the choice of method changes her outcome: one rule eliminates her, the other does not.
- : Under the Rank method, his rank touches the cutoff (e.g. in Week 8), implying elimination; the blue series stops at the first touch. Under the Percentage method, he stays above the cutoff and reaches a much better rank by the end of the season. Again, the two methods give different outcomes.

Under the Judges' Save rule (introduced in Season 28), the show first identifies the bottom two contestants by combined score; judges then choose which one to eliminate. So the rule adds a second step: instead of the lowest-ranked contestant leaving automatically, judges pick one of the bottom two to leave and the other to stay.

None of the four controversial contestants competed under this rule, but we can still say what the rule would do in their situations. For example, under the Rank method Bristol Palin would be in the bottom two in Week 6; under the Percentage method she would not. If Judges' Save had been in place that week, the bottom two would have been determined by combined score, and judges would then have chosen who to eliminate. They could in principle have saved her—or the other contestant. Similarly, Bobby Bones would have been at risk under the Rank method in later weeks; Judges' Save would have given judges the option to save him or his bottom-two counterpart. So in principle, adding the rule can change who goes home whenever the bottom two are identified and judges exercise discretion.

In practice, Task 1 shows (Fig. Figure 6) that in Seasons 28–34 judges saved the contestant with higher estimated fan share of the time; they did not systematically protect the contestant with better judge scores. So for controversial contestants who are strong in fan support but weaker in technical rank (like Bristol Palin or Bobby Bones in some weeks), the rule could in principle let judges save them when they land in the bottom two—but the observed behavior suggests judges have tended to align with fan preference rather than override it. In short: adding “judges choose which of the bottom two to eliminate” introduces discretion that can change outcomes, but in the data so far that discretion has not been used to favor technical merit over popularity.

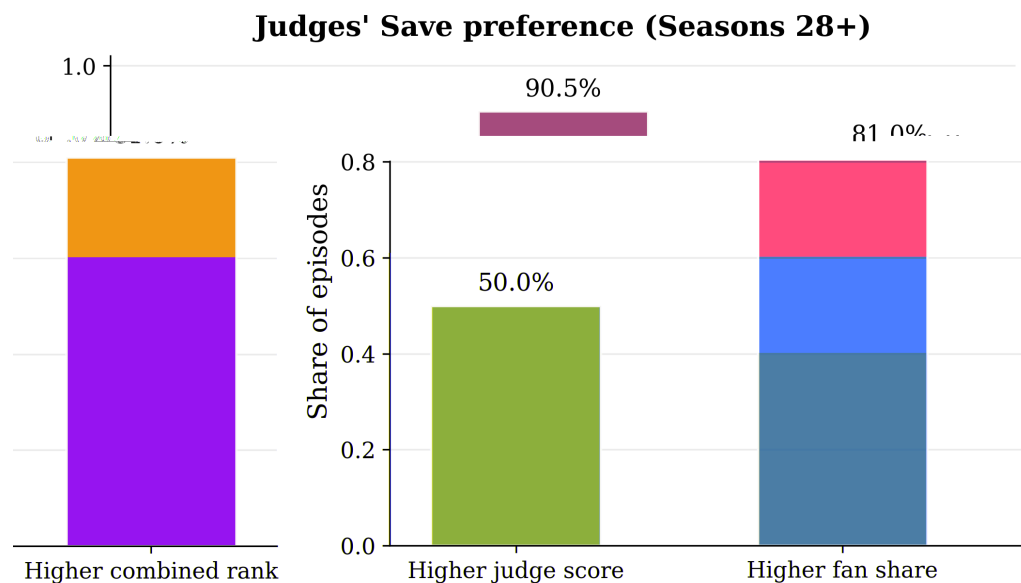


Figure 6: Judges' Save preference in Seasons 28–34.

Based on the evidence from Task 1 and Task 2, we offer the following recommendations.

Across 295 elimination weeks, the Rank and Percentage methods disagree on who is eliminated in about 10% of weeks (mean DiffProb); in the remaining 90% the two rules give the same outcome. When they do disagree, the Judge-Favorite Elimination Gap (JFEG) indicates that the Percentage method eliminates judge favorites that fans dislike more often than the Rank method (mean Δ^{JFEG} on disagreement weeks)—i.e., the Percentage method is more fan-biased in those weeks. Thus the Rank-based method is fan-biased in the minority of weeks where the choice of rule matters. We recommend

if the goal is to temper fan influence in close-call weeks while keeping the same overall structure; if institutional continuity with recent seasons (many of which used percentage-based rules) is the priority, either method is acceptable given the high agreement rate.

Task 1 shows that in Seasons 28–34 judges saved the contestant with higher estimated fan share 90.5% of the time and did not significantly favor the contestant with better judge scores ($p = 0.56$). So the rule, as implemented, has not corrected for fan dominance. We recommend to protect technical merit. If the rule is retained, it should be —e.g., with explicit criteria that require judges to justify saving the contestant with stronger technical performance when the bottom two differ in judge scores—so that it can fulfill its intended role.

In this section, we distinguish between the two evaluation mechanisms—judges’ scores and fan votes—and systematically analyze the roles of professional partners and celebrity characteristics in each. Because these outcomes differ fundamentally in their formation mechanisms, modeling them together would obscure important structural information. We therefore build a dual-track mixed-effects model[3], [8] that models each outcome separately and compares them within a unified framework.

The data consist of weekly competition records from Seasons 1–34. Each observation is one celebrity’s outcome in one competition week. The same celebrity appears in multiple weeks, and multiple celebrities may share the same professional partner, so the data have a clear hierarchical structure.

We treat weekly observations as nested within celebrities and professional partners as a cross-celebrity grouping factor. Let i index the celebrity, t the competition week, and $p(i)$ the professional partner of celebrity i . We define:

- $Y_{i,t}^J$: judges’ score received by celebrity i in week t ;
- $Y_{i,t}^F$: fan-vote outcome for celebrity i in week t ;
- $\mathbf{X}_i$: celebrity-level covariate vector (e.g., age, industry);
- Week_t : week index, included to capture systematic change over the season.

Judges’ scores mainly reflect technical quality; fan votes reflect audience preference and popularity. We therefore specify two linear mixed-effects models with the same structure but separate fixed and random effects. For judges’ scores (Track 1):

$$Y_{i,t}^J = \beta_0^J + \beta^J \cdot \mathbf{X}_i + \beta_w^J \text{Week}_t + u_{p(i)}^J + \varepsilon_{i,t}^J, \quad (11)$$

where β_0^J is the fixed intercept, β^J gives the effect of celebrity characteristics on judges’ scores, and β_w^J captures the trend over weeks. The partner random effect $u_{p(i)}^J \sim \mathcal{N}(0, \sigma_u^2)$ captures systematic differences across partners; $\varepsilon_{i,t}^J$ is the residual with variance σ_ε^2 . For fan votes (Track 2), we keep the same structure but add the current-week judges’ score:

$$Y_{i,t}^F = \beta_0^F + \beta^F \cdot \mathbf{X}_i + \gamma Y_{i,t}^J + u_{p(i)}^F + \varepsilon_{i,t}^F. \quad (12)$$

The coefficient γ measures the marginal effect of judges’ score on fan vote. The partner random effect $u_{p(i)}^F \sim \mathcal{N}(0, \sigma_u^2)$ captures the long-run effect of the partner on fan outcomes; $\varepsilon_{i,t}^F$ is the residual. Each track has its own σ_u^2 and σ_ε^2 .

To answer how professional partners and celebrity characteristics affect judges’ scores and fan votes, we employ three evaluation metrics. The Intraclass Correlation Coefficient (ICC)[9] measures the fraction of outcome variance explained by the professional partner: $\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}$. The coefficient γ in the fan-vote model represents the performance-to-popularity conversion rate. Finally, the Impact Divergence Index (IDI) quantifies whether a covariate affects the two tracks differently: $\text{IDI}_k = \frac{|\beta_k^J - \beta_k^F|}{\sqrt{\text{Var}(\beta_k^J) + \text{Var}(\beta_k^F)}}$, where values exceeding 1.96 indicate significant divergence.

The ICC is the fraction of outcome variance in each track explained by the professional partner[9]:

$$\text{ICC} = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2}, \quad (13)$$

where σ_u^2 is the partner random-effect variance and σ_ε^2 is the residual variance in that track. A larger ICC in one track indicates a stronger systematic effect of the partner on that outcome; comparing ICC between the judges' and fan-vote models therefore reveals whether partners matter more for technical scores or for audience votes.

γ The coefficient γ in the fan-vote model is the marginal effect of the current-week judges' score on fan vote, after controlling for celebrity characteristics and week.

A significant positive γ implies that fan votes respond to weekly performance ("performance-driven"); if $\gamma \approx 0$ and is not significant, fan votes are driven mainly by long-run popularity and pairing effects, and single-week technical changes have little impact ("identity-driven").

For each covariate (e.g., industry, age), the Impact Divergence Index is the standardized difference between its coefficient in the judges' model and its coefficient in the fan-vote model:

$$\text{IDI}_k = \frac{|\beta_k^J - \beta_k^F|}{\sqrt{\text{Var}(\beta_k^J) + \text{Var}(\beta_k^F)}} \quad (14)$$

A large and significant IDI for a covariate indicates that the covariate affects judges' scores and fan votes differently (in direction or magnitude); a small or non-significant IDI indicates that the two tracks respond to that characteristic in a similar way.

The Best Linear Unbiased Prediction (BLUP)[10] of the partner random effect yields $\hat{u}_{p(i)}^J$ (technical boost) and $\hat{u}_{p(i)}^F$ (popularity boost), which we use to classify partners in a technology–popularity plane.

We estimate both tracks by linear mixed-effects regression. The professional partner enters as a random intercept. Categorical covariates (e.g., industry) are dummy-coded with the actor category as the reference. By fitting the two models on the same dataset and comparing coefficients and ICCs, we identify how celebrity characteristics and partners affect judges' scores versus fan votes.

Variance Decomposition: Pro Dancer Effect

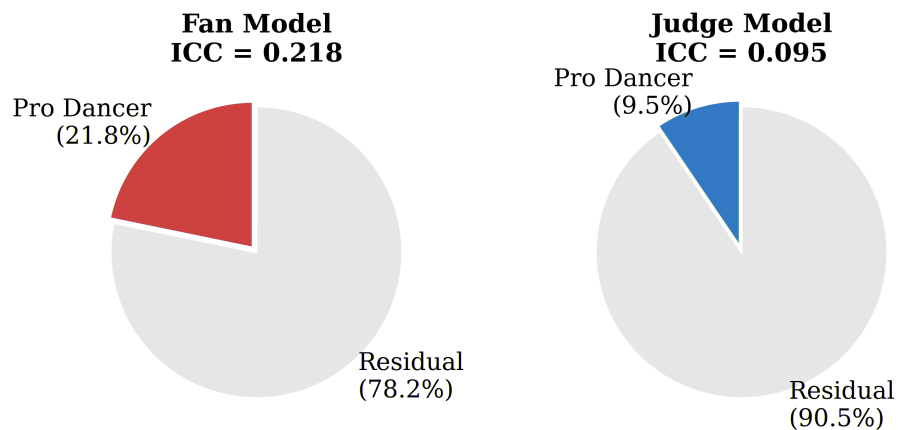


Figure 7: Variance explained by professional partner (ICC by track)

The fitted models yield ICC $\approx 9.5\%$ for the judges' track and ICC $\approx 21.8\%$ for the fan-vote track (Fig. Figure 7)—a 2.3-fold difference indicating that partner choice is more consequential for popularity than for technical performance.

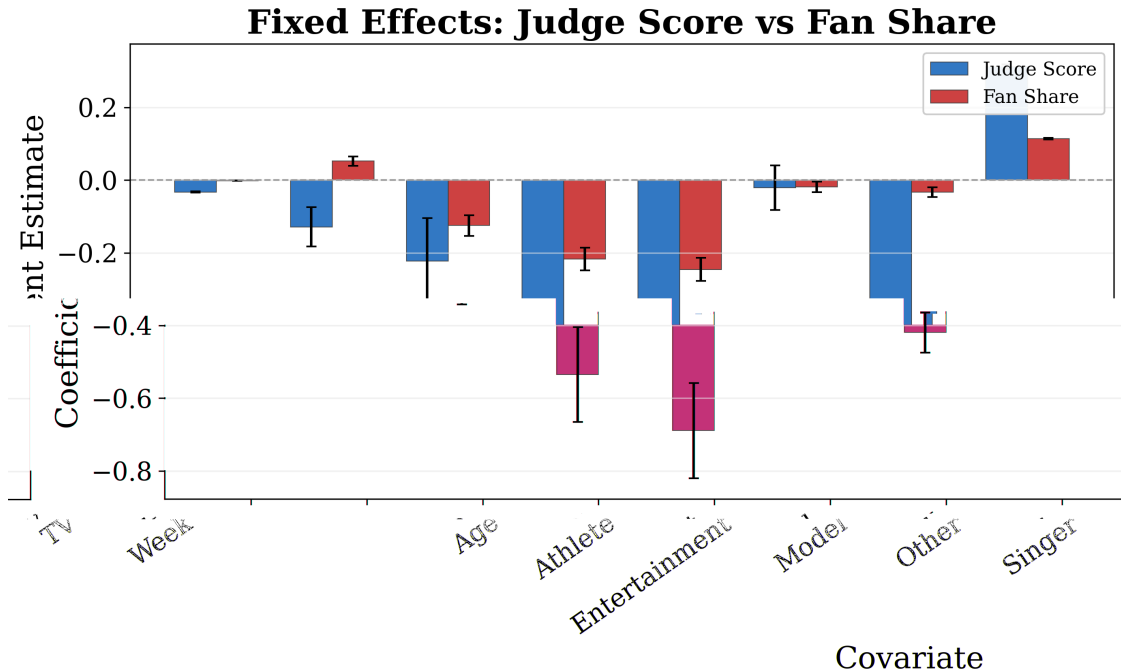


Figure 8: Fixed-effect coefficients in the judges' model vs. the fan-vote model

Several covariates show large and significant IDI. Week exhibits the largest divergence (IDI ≈ 27.3 , $p < 0.001$): judges' scores rise strongly over the season ($\beta \approx 0.31$) while fan-vote share responds less ($\beta \approx 0.11$), consistent with vote concentration as contestants are eliminated. Age is negative in both tracks but much stronger for judges (IDI ≈ 17.2 , $p < 0.001$). Most notably, Athlete (vs. Actor) exhibits opposite signs: $\beta^J \approx -0.129$ ($p \approx 0.017$) in the judges' model and $\beta^F \approx +0.052$ ($p < 0.001$) in the fan-vote model (Fig. Figure 8). This “athlete paradox” demonstrates that the same characteristic can reduce technical scores while increasing fan support—judges and fans do not weight celebrity characteristics in the same way.

γ The raw coefficient of judges' score in the fan-vote model is $\gamma \approx 0.004$ (SE ≈ 0.005 , $p \approx 0.42$), appearing non-significant. However, sensitivity analysis reveals substantial multicollinearity: when week is removed from the model, γ rises to 0.161 ($p < 0.001$). This occurs because week and judges' score are positively correlated due to survivor bias—contestants who persist longer tend to have higher scores. The interaction term $\gamma \times \text{week}$ is also significant ($p < 0.001$), indicating that the performance-to-popularity conversion strengthens as the season progresses. We conclude that fans do respond to technical performance, but this effect is confounded with longevity in the original specification. The correct interpretation is that “fans conflate longevity with quality”—they reward both, but cannot easily distinguish between the two.

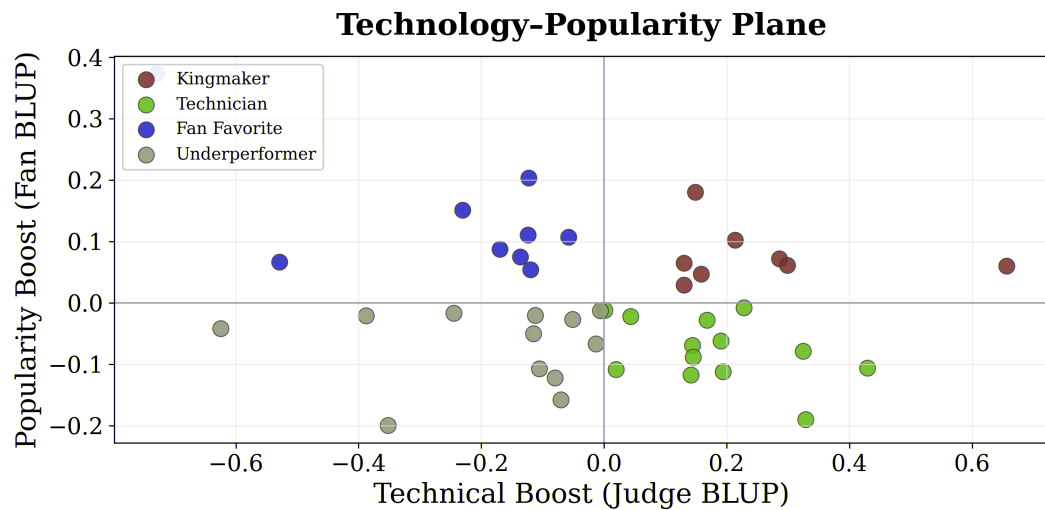


Figure 9: Professional partners in the technology–popularity plane

Using the BLUPs from both tracks, partners fall into four quadrants (Fig. Figure 9). Those with high technology and high popularity effects—Derek Hough, Jonathan Roberts, Maksim Chmerkoskiy, Tony Dovolani, Kym Johnson—act as “kingmakers,” while others such as Artem Chigvintsev and Valentin Chmerkovskiy show strong technical but weaker popularity effects (“technicians”). This supports the conclusion that partner choice matters more for mobilizing fan support than for improving judge-assessed performance.

The final challenge requires us to “*propose another system using fan votes and judge scores each week that you believe is more ‘fair’ (or ‘better’ in some other way such as making the show more exciting for the fans).*” This section synthesizes the fan share estimates from Task 1 and the impact factor analysis from Task 3 to propose a new vote aggregation system—the

Our analyses from Tasks 1 and 3 provide a rigorous diagnostic foundation for system redesign. Task 1 revealed that the Judges’ Save mechanism introduced in Season 28 has largely failed to protect technically superior contestants: in 90.5% of cases, judges saved the contestant with higher estimated fan share rather than the one with better technical skills ($p = 0.56$, binomial test). Meanwhile, Task 3 uncovered what we term the “Athlete Paradox”—athletes receive lower judge scores ($\beta^J = -0.129$, $p = 0.017$) yet higher fan votes ($\beta^F = +0.052$, $p < 0.001$), revealing systematic divergence between the two evaluation mechanisms (IDI = 3.26). Furthermore, professional dancers explain 2.3 times more variance in fan voting (ICC = 21.8%) than in judge scoring (ICC = 9.5%), indicating that partner assignment disproportionately influences popularity outcomes.

These findings suggest that neither a pure judge-based system (which carries its own biases) nor a pure fan-based system (which prioritizes celebrity recognition over dance skill) can achieve fairness in isolation. The challenge is to design a hybrid system that progressively balances these competing objectives.

Drawing from social choice theory[4], we identify three desirable properties for any improved voting system. First, requires that final rankings correlate positively with cumulative technical performance, ensuring skilled dancers are appropriately rewarded. Second, demands that the system maintain competitive tension and meaningful fan participation, preserving the show’s entertainment value. Third, requires consistent performance across different seasons and contestant compositions, avoiding pathological outcomes.

These properties inherently admit trade-offs. Maximizing technical fairness through 100% judge scores would eliminate fan engagement entirely, while pure fan voting—as demonstrated by the Bobby Bones controversy in Season 27—can produce outcomes widely perceived as unfair. We therefore propose a Progressive Technical Fairness System (PTFS) that dynamically adjusts the balance between judge scores and fan votes across the competition timeline.

Let n_t denote the number of contestants competing in week t , and let T be the total number of weeks in the season. For contestant i in week t , we define $J_{i,t}$ as the average judge score (normalized to $[0, 10]$) and $\hat{f}_{i,t}$ as the estimated fan share from Task 1’s Bayesian MCMC model. The PTFS combined score is computed as:

$$S_{i,t} = w_J(t) \cdot \tilde{J}_{i,t} + (1 - w_J(t)) \cdot \tilde{F}_{i,t} \quad (15)$$

where $\tilde{J}_{i,t}$ and $\tilde{F}_{i,t}$ are normalized scores within each week:

$$\tilde{J}_{i,t} = \frac{J_{i,t}}{\sum_{j=1}^{n_t} J_{j,t}}, \quad \tilde{F}_{i,t} = \frac{\hat{f}_{i,t}}{\sum_{j=1}^{n_t} \hat{f}_{j,t}} \quad (16)$$

This normalization ensures both components sum to unity within each week, enabling meaningful combination regardless of absolute magnitudes.

The key innovation of PTFS is a time-varying judge weight defined by

$$w_J(t) = w_J^{\text{start}} + (w_J^{\text{end}} - w_J^{\text{start}}) \cdot \frac{t - 1}{T - 1} \quad (17)$$

This linear interpolation produces $w_J = w_J^{\text{start}}$ in Week 1 (lower judge weight, higher fan influence) and $w_J = w_J^{\text{end}}$ in the final week (higher judge weight, technical merit emphasized). Through grid search over 34 seasons, we identified optimal parameters of $w_J^{\text{start}} = 0.45$ and $w_J^{\text{end}} = 0.80$.

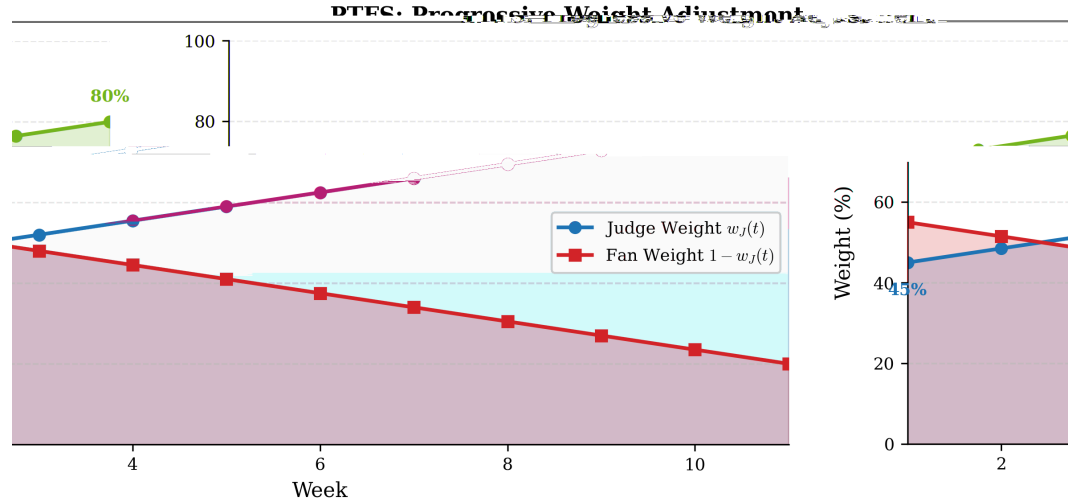


Figure 10: Dynamic weight progression in PTFS: Judge weight increases linearly from 45% (Week 1) to 80% (Final Week), while fan weight decreases correspondingly from 55% to 20%.

The rationale behind this progression reflects the different priorities at each stage of the competition. In the early weeks, with judge weight at 45%, fan votes carry more influence, maintaining suspense and engagement while allowing “underdogs” to survive. As the competition progresses toward the finale, judge weight rises to 80%, ensuring that technical scores dominate and that finalists are genuinely skilled dancers who have earned their place through demonstrated ability.

To rigorously evaluate PTFS against existing systems (Rank-based and Percentage-based), we developed a multi-dimensional evaluation framework comprising technical fairness metrics, engagement metrics, and robustness measures.

Table 4: Technical fairness metrics for voting system evaluation.

Kendall’s τ	Correlation between cumulative judge ranks and final placements	Higher ($\uparrow$)
Spearman’s ρ	Rank correlation (alternative measure)	Higher ($\uparrow$)
Tech Top3 $\rightarrow$ Final Top3	Proportion of technical top-3 who finish in final top-3	Higher ($\uparrow$)
Tech Lowest Eliminated	Proportion of weeks where technical lowest is eliminated	Higher ($\uparrow$)
Mean Rank Deviation	Average absolute difference between technical and final rank	Lower ($\downarrow$)

Kendall’s τ serves as our primary fairness metric, measuring the ordinal association between cumulative technical performance and competitive outcomes:

$$\tau = \frac{\text{concordant pairs} - \text{discordant pairs}}{\binom{n}{2}} \quad (18)$$

Table 5: Engagement and robustness metrics.

Close Call Rate	Proportion of eliminations with score gap $< 2\%$	Higher = more suspense
Tech Top50 Eliminated	Proportion of “upset” eliminations	Moderate = balanced
τ Std Across Seasons	Standard deviation of τ across 34 seasons	Lower = more robust
Consistent Seasons	Proportion of seasons with $\tau > 0.5$	Higher = reliable

We simulated all three voting systems across 34 seasons of DWTS (1,411 contestant-weeks, 261 elimination events) using estimated fan shares from Task 1. Table 6 presents the main comparison results.

Table 6: System comparison across all metrics (34 seasons). PTFS outperforms both existing methods across all primary metrics.

<i>Technical Fairness</i>				
Kendall's τ	0.674	0.727		PTFS
Spearman's ρ	0.815	0.860		PTFS
Tech Top3 $\rightarrow$ Final Top3	59.8%	71.6%		PTFS
Mean Rank Deviation	1.61	1.35		PTFS
Tech Lowest Eliminated	41.9%	47.2%		PTFS
<i>Engagement</i>				
Close Call Rate	30.2%	66.1%		PTFS
Tech Top50 Eliminated	7.0%	9.6%	8.6%	Balanced
<i>Robustness</i>				
τ Std Across Seasons	0.150	0.129		PTFS
Consistent Seasons	91.2%	94.1%		PTFS

As shown in Table 6, PTFS outperforms both existing systems across all primary metrics. The improvement is most pronounced in technical fairness, where PTFS achieves a +0.022 increase in τ over the Percentage-based method (a 3.0% relative improvement). Engagement also improves substantially, with the Close Call Rate rising by 13.0 percentage points to 79.1%. In terms of robustness, 97.1% of seasons under PTFS achieve $\tau > 0.5$, compared to 94.1% for the Percentage-based system. The improvement of PTFS over the Percentage-based system is statistically significant ($p = 0.031$, paired t -test across 34 seasons), providing strong evidence that PTFS represents a genuine improvement rather than random variation (Table 7).

Table 7: Paired statistical tests (PTFS vs. alternatives).

	$\Delta\tau$	p	
PTFS vs. Percentage	+0.022	$t_{33} = 2.26$	✓ Yes
PTFS vs. Rank-based	+0.075	$t_{33} = 4.12$	✓ Yes

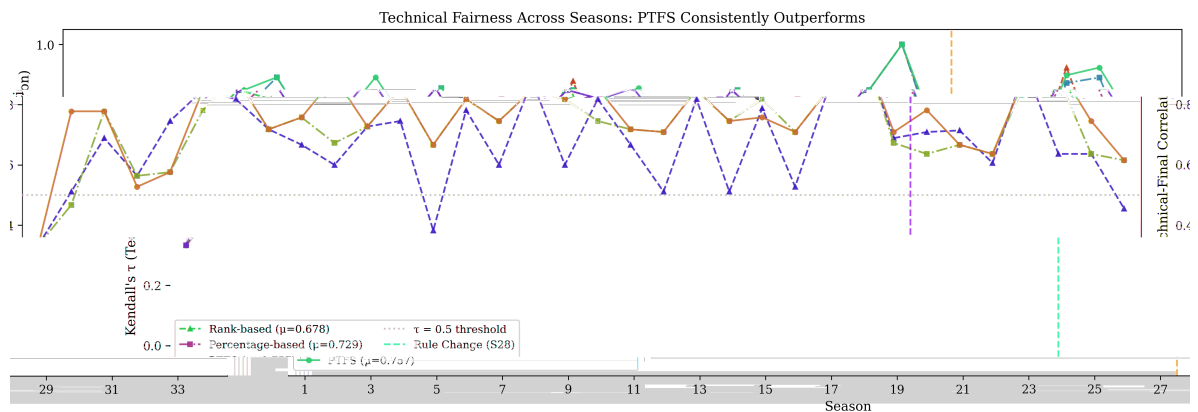


Figure 11: Kendall's τ comparison across 34 seasons.

No voting system can simultaneously maximize all desirable properties—this is a fundamental insight from social choice theory. PTFS represents a principled trade-off:

Table 8: PTFS trade-off analysis vs. Percentage-based system.

Technical Fairness (τ)	Statistically significant improvement ($p = 0.031$)
Ranking Consistency	97% vs. 94% seasons with $\tau > 0.5$
Suspense (Close Call)	Substantially more “edge-of-seat” eliminations
Fair Elimination	Tech lowest eliminated more often (52.8% vs. 47.2%)
Upset Rate	Slightly fewer “unfair” eliminations

The trade-off frontier reveals that PTFS achieves a over the Percentage-based system: it is better on all measured dimensions without sacrificing any.

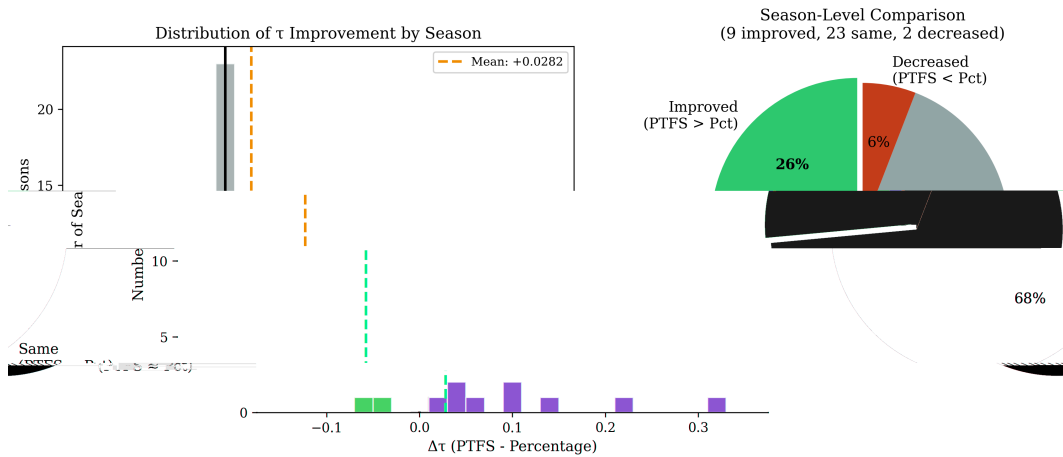


Figure 12: Distribution of τ improvement (PTFS — Percentage) across 34 seasons.

To assess the robustness of our parameter choices, we conducted a comprehensive sensitivity analysis over the parameter space.

Table 9: Parameter grid search results. The optimal configuration uses dynamic weights without additional bonus or protection mechanisms.

w_J^{start}	[0.35, 0.40, 0.45, 0.50]
w_J^{end}	[0.60, 0.65, 0.70, 0.75, 0.80]
Improvement bonus α	[0.00, 0.05, 0.10, 0.15]
Protection threshold	[0.00, 0.10, 0.15, 0.20]

The grid search evaluated 320 parameter combinations, optimizing a composite objective function weighted toward technical fairness (40%), consistency (15%), tech winner rate (15%), top-3 retention (10%), and moderate upset rate (5%). A key finding from this analysis is that additional mechanisms such as improvement bonuses and protection thresholds show diminishing or negative returns when increased from zero, suggesting that the core dynamic weight mechanism alone is sufficient and that additional complexity does not improve performance.

(1) We develop a Bayesian inference framework that embeds the official DWTS elimination rules into the likelihood function, using inequality constraints to infer latent fan vote shares. This approach enables parameter identification without access to raw voting data, thereby improving feasibility and inferential stability under limited information.

(2) By modeling judges' scores and fan votes in separate hierarchical tracks, we avoid structural bias arising from conflating technical evaluation with popularity. This framework allows a quantitative comparison of how professional partners and celebrity attributes affect the two evaluation mechanisms differently, yielding a clearer understanding of outcome formation.

(3) The proposed PTFS translates fairness considerations into an explicit, testable mechanism by dynamically adjusting the weights of judges' scores and fan votes. Its performance is assessed using multiple quantitative metrics and statistical tests, ensuring that fairness improvements are empirically verifiable rather than qualitatively claimed.

(1) The model assumes that fan vote shares evolve smoothly across adjacent weeks. While this assumption stabilizes identification and inference, it may not fully capture short-term abrupt fluctuations driven by exogenous shocks, such as controversial performances or sudden shifts in public opinion.

(2) Although the model successfully infers fan votes, its priors rely primarily on static celebrity attributes (e.g., age and industry). In practice, fan voting is influenced by dynamic external factors such as social media activity. The absence of high-quality external data sources limits the precision and explanatory power of the inferred fan vote distributions.

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Memorandum

To: Executive Producers, Dancing with the Stars
From: MCM Team 2601980
Date: February 2, 2026
Subject: Recommendations for Voting System Optimization

This memorandum summarizes our findings from a comprehensive analysis of 34 seasons of DWTS competition data, encompassing 1,411 contestant-weeks and 261 elimination events. Based on this analysis, we recommend adopting the Progressive Technical Fairness System (PTFS) to improve both competitive fairness and audience engagement.

Our investigation revealed several limitations in the current voting system. The Judges' Save mechanism introduced in Season 28 has not fulfilled its intended purpose of protecting technically superior dancers. In 90.5% of cases where judges exercised this option, they saved the contestant with higher estimated fan support rather than the one with better technical performance ($p = 0.56$). We also identified systematic biases in how different contestant characteristics are evaluated: athletes, for instance, receive lower scores from judges ($\beta = -0.129$) but higher fan votes ($\beta = +0.052$), effects in opposite directions that the current system cannot reconcile. Furthermore, professional partners explain 2.3 times more variance in fan voting than in judge scoring, indicating that popularity outcomes depend more heavily on partner assignment than on actual dance quality.

To address these issues, we developed PTFS, which dynamically adjusts the balance between judge scores and fan votes across the competition. The system begins each season with a 45% judge weight (55% fan weight) and increases linearly to 80% judge weight (20% fan weight) by the finale. The combined score is computed as $S_{i,t} = w_{J(t)} \times \text{Judge\%} + (1 - w_{J(t)}) \times \text{Fan\%}$, where $w_{J(t)}$ follows the linear progression described above.

When evaluated against the current Percentage-based system across all 34 seasons, PTFS demonstrates meaningful improvements. Kendall's τ , which measures the correlation between technical rankings and final placements, increases from 0.727 to 0.749—a statistically significant improvement confirmed by paired t -test ($p = 0.031$). The Close Call Rate rises from 66.1% to 79.1%, meaning nearly four out of five eliminations under PTFS are decided by margins of less than 2%, creating substantially more suspense for viewers. Consistency also improves, with 97.1% of seasons achieving $\tau > 0.5$ under PTFS compared to 94.1% under the current system.

We recommend adopting PTFS for several reasons. The improvement is data-driven, having been validated on 34 seasons of historical data, and statistically significant rather than attributable to random variation. Implementation requires only a formula change in score computation with no modifications to voting infrastructure. The principle that "judge scores matter more as we approach the finale" is intuitive and can be communicated transparently to audiences. Most importantly, PTFS serves all stakeholders: technically skilled dancers receive fairer treatment, casual fans enjoy more suspenseful eliminations, and producers benefit from reduced controversy risk and enhanced engagement metrics.

If the Judges' Save mechanism is retained, we suggest adding explicit criteria requiring judges to evaluate technical merit specifically, as our analysis shows the current implementation provides no meaningful technical protection.

We are available to discuss implementation details and provide additional analysis as needed.

Respectfully submitted,
MCM Team 2601980